

Chapter 23

Fields of a moving point charge

23.1 The Liénard–Wiechert potential

The current of a point charge q moving along a worldline $y^\mu(\tau)$ is

$$j^\mu(y) = q \int u^\mu(\tau) \delta^4(y - y(\tau)) d\tau \quad (23.1)$$

In general, we expect the charge to accelerate – leading to the *Liénard–Wiechert* fields. Using the retarded Green’s function, the potential becomes

$$\begin{aligned} A^\mu &= \frac{2q}{c} \int d^4y \int d\tau u^\mu(\tau) \theta(x^0 - y^0) \delta((x - y)^2) \delta^4(y - y(\tau)) \\ &= \frac{2q}{c} \int d\tau u^\mu(\tau) \theta(x^0 - y^0(\tau)) \delta((x - y(\tau))^2) \\ &= \frac{2q}{c} \int d\tau u^\mu(\tau) \theta(x^0 - y^0(\tau)) \delta(f(\tau)) \end{aligned} \quad (23.2)$$

where

$$f(\tau) = (x - y(\tau))^2 = (x^0 - y^0(\tau))^2 - |\vec{x} - \vec{y}(\tau)|^2, \quad f'(\tau) = -2(x^\mu - y^\mu(\tau)) u_\mu(\tau) \quad (23.3)$$

The delta function localizes onto the zeroes of f

$$\theta(x^0 - y^0(\tau)) \delta(f(\tau)) = \theta(x^0 - y^0(\tau)) \left[\frac{\delta(\tau - \tau_+)}{f'(\tau_+)} + \frac{\delta(\tau - \tau_-)}{f'(\tau_-)} \right] \quad (23.4)$$

where $\tau_\pm = \tau_\pm(x)$ are the zeroes of f , as depicted in the Figure. The step

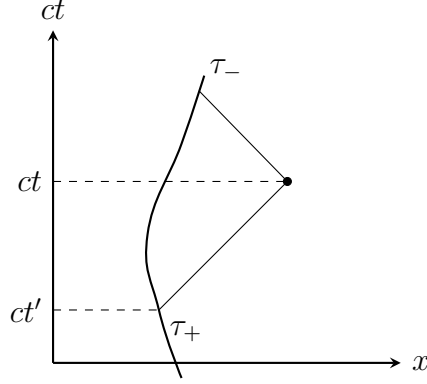


Figure 23.1: The worldline of the charge (curve) and an observation point at time ct : the light cone through the observation point intersects the worldline at proper times $\tau_{\pm}$; the step function keeps only the retarded intersection τ_+ , at time ct' .

function then retains only the retarded zero τ_+

$$\theta(x^0 - y^0(\tau)) \delta(f(\tau)) = \frac{\delta(\tau - \tau_+)}{f'(\tau_+)} = \frac{\delta(\tau - \tau_+)}{2(x - y(\tau_+)) \cdot u(\tau_+)} \quad (23.5)$$

At τ_+ , we have

$$t - t' = \frac{1}{c} |\vec{x} - \vec{y}(t')| \quad (23.6)$$

where t' is the *retarded time*. We then arrive at the Liénard–Wiechert potential

$$A^\mu = \frac{q}{c} \frac{u^\mu(\tau_+)}{(x - y(\tau_+)) \cdot u(\tau_+)} \quad (23.7)$$

23.2 The field strength

Write

$$L^\mu \equiv x^\mu - y^\mu(\tau_+) \quad , \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (23.8)$$

Then

$$F^{\mu\nu} = \frac{q}{c(uL)^3} [L^\mu u^\nu c^2 + L^\mu ((uL) a^\nu - (aL) u^\nu)] - (\mu \leftrightarrow \nu) \quad (23.9)$$

where

$$a^\mu = \frac{du^\mu}{d\tau} \quad (23.10)$$

is the four-acceleration, with the shorthand $(uL) \equiv u^\mu L_\mu$, $(aL) \equiv a^\mu L_\mu$, and so on.

To compute $F^{\mu\nu}$: the retarded proper time $\tau_+(x)$ is defined implicitly by

$$(x - y(\tau_+))^2 = 0 \quad (23.11)$$

Applying ∂_ν ,

$$2(x_\mu - y_\mu(\tau_+))(\delta^\mu_\nu - u^\mu \partial_\nu \tau_+) = 0 \Rightarrow (x_\nu - y_\nu(\tau_+)) = (x_\mu - y_\mu(\tau_+)) u^\mu \partial_\nu \tau_+ \quad (23.12)$$

so that

$$\partial_\nu \tau_+ = \frac{x_\nu - y_\nu(\tau_+)}{(x - y) \cdot u} = \frac{L_\nu}{(Lu)} \quad (23.13)$$

Next,

$$\begin{aligned} \partial_\nu(uL) &= \partial_\nu u^\mu L_\mu + u^\mu \partial_\nu L_\mu \\ &= a^\mu L_\mu \partial_\nu \tau_+ + u_\nu - u^\mu u_\mu \partial_\nu \tau_+ \\ &= ((aL) - c^2) \partial_\nu \tau_+ + u_\nu = \frac{((aL) - c^2) L_\nu}{(Lu)} + u_\nu \end{aligned} \quad (23.14)$$

which gives

$$\begin{aligned} \partial_{[\mu} A_{\nu]} &= \frac{2q}{c} \frac{a_{[\mu} L_{\nu]}}{(Lu)^2} - \frac{2q}{c} \frac{u_{[\mu} L_{\nu]} ((aL) - c^2)}{(uL)^3} \\ &= \frac{2q}{c(uL)^3} [a_{[\mu} L_{\nu]} (uL) - u_{[\mu} L_{\nu]} ((aL) - c^2)] \end{aligned} \quad (23.15)$$

reproducing (23.9).

We can write this instead as

$$\begin{aligned} F^{\mu\nu} &= \frac{q}{(um)^3 R^2} (m^\mu u^\nu - m^\nu u^\mu) \\ &\quad + \frac{q}{(um)^3 R} (m^\mu ((um) a^\nu - (am) u^\nu) - (\mu \leftrightarrow \nu)) \end{aligned} \quad (23.16)$$

where we defined

$$R \equiv L^0 = |\vec{x} - \vec{y}(\tau_+)| \quad , \quad m^\mu \equiv \frac{L^\mu}{R} \quad , \quad m_\mu m^\mu = 0 \quad , \quad m^0 = 1 \quad (23.17)$$

that is

$$\vec{m} = \frac{\vec{x} - \vec{y}(\tau)}{|\vec{x} - \vec{y}(\tau)|} \quad , \quad |\vec{m}| = 1 \quad (23.18)$$

The first term in (23.16) scales as $1/R^2$ – the Coulomb-like piece; the second scales as $1/R$ and is due to acceleration.

23.3 Uniform motion

For uniform motion, $y^\mu(\tau) = u^\mu \tau$ with constant u^μ , so that

$$L^\mu = x^\mu - y^\mu(\tau_+) = x^\mu - u^\mu \tau_+ \Rightarrow (uL) = (ux) - c^2 \tau_+ \quad (23.19)$$

The retarded condition $f(\tau_+) = 0$ reads

$$(x - \tau_+ u)^2 = x^2 + \tau_+^2 c^2 - 2 \tau_+ (ux) = 0 \quad (23.20)$$

giving

$$\tau_+ = \frac{(ux)}{c^2} - \frac{1}{c^2} \sqrt{(ux)^2 - x^2 c^2} \Rightarrow (uL) = \sqrt{(ux)^2 - x^2 c^2} > 0 \quad (23.21)$$

and hence

$$A^\mu = \frac{q}{c} \frac{u^\mu(\tau_+)}{\sqrt{(ux)^2 - x^2 c^2}} [?] \quad (23.22)$$

As a check, take $u^\mu = (c, 0, 0, 0)$: then

$$(ux)^2 - x^2 c^2 = c^4 t^2 - c^4 t^2 + c^2 |\vec{x}|^2 \Rightarrow A^0 = \frac{q}{|\vec{x}|} [?] \quad (23.23)$$

– the Coulomb potential, as expected. The field strength is

$$F^{\mu\nu} = \frac{q}{c} \frac{x^\mu u^\nu - x^\nu u^\mu}{((ux)^2 - x^2 c^2)^{3/2}} [?] \quad (23.24)$$

Define $\vec{R} = \vec{x} - \vec{v} t$; one finds

$$(ux)^2 - x^2 = \frac{R^2 + (\vec{v} \cdot \vec{R})^2 - v^2 R^2}{1 - v^2/c^2} \quad (23.25)$$

and the electric field becomes

$$\vec{E} = \frac{1 - \frac{v^2}{c^2}}{(1 - \frac{v^2}{c^2} \sin^2 \theta)^{3/2}} \frac{q \vec{R}}{R^3} \quad (23.26)$$

where θ is the angle between $\vec{v}$ and $\vec{R}$: the field is radial but *not* spherically symmetric. Comparing with the Coulomb value E_0 ,

$$\theta = \frac{\pi}{2} : E_\perp > E_0 \quad , \quad \theta = 0, \pi : E_\parallel < E_0 \quad (23.27)$$

– the field lines focus towards $\theta = \pm\pi/2$, as depicted in the Figure. The magnetic field lines circle around the direction of motion, with

$$\vec{B} = \frac{\vec{v}}{c} \times \vec{E} \quad (23.28)$$

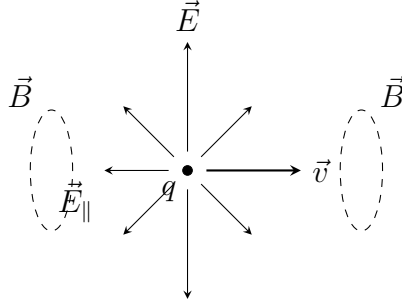


Figure 23.2: Field of a uniformly moving charge: $\vec{E}$ is enhanced perpendicular to the motion and reduced along it, while the $\vec{B}$ field lines circle around the direction of motion.

23.4 Velocity and acceleration fields

Coming back to the general case, split the fields as

$$\vec{E} = \vec{E}_v + \vec{E}_a \quad , \quad \vec{B} = \vec{B}_v + \vec{B}_a \quad (23.29)$$

The velocity fields are

$$\vec{E}_v = \frac{2q[?]}{R^2} \frac{\left(1 - \frac{v^2}{c^2}\right) \left(\vec{m} - \frac{\vec{v}}{c}\right)}{\left(1 - \frac{\vec{v} \cdot \vec{m}}{c}\right)^3} \quad , \quad \vec{B}_v = \vec{m} \times \vec{E}_v \quad (23.30)$$